

## Solution Requirements

|                      |                                                    |
|----------------------|----------------------------------------------------|
| <b>Date</b>          | 15 July 2026                                       |
| <b>Team ID</b>       | SWUID2026234324                                    |
| <b>Project Name</b>  | VideoScriptAI – AI Powered Video Content Generator |
| <b>Maximum Marks</b> | 4 Marks                                            |

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category    | Requirement Description                                                                                                         | Priority (High/Medium/Low) |
|------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1    | Authentication          | <i>The application securely authenticates API requests using the Groq API key stored in the .env file.</i>                      | High                       |
| 2    | Authorization levels    | <i>The application currently supports a single user role with access to all content generation features.</i>                    | Medium                     |
| 3    | External interfaces     | <i>Integrates with the Groq API (LLaMA 3.3-70B Versatile) to generate AI-powered video content.</i>                             | High                       |
| 4    | Transactions processing | <i>User input is validated and processed before sending requests to the Groq API. Generated responses are returned as JSON.</i> | High                       |

|   |                                   |                                                                                                                                                                    |        |
|---|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 5 | Reporting                         | <i>Displays generated Hook, Script, Scene Suggestions, and CTA on the web interface for review and use.</i>                                                        | Medium |
| 6 | Business rules, etc.              | <i>Product name, description, target audience, and campaign goal are mandatory before content generation. Duration is optional.</i>                                | High   |
| 7 | Compliance to laws or regulations | <i>API credentials are securely stored using environment variables. The application follows secure coding practices and avoids exposing sensitive information.</i> | Medium |
| 8 | Other                             | <i>[Add any additional functional requirements here]</i>                                                                                                           | High   |

**Step 2: Non-Functional Requirements (NFR)**

|   |                     |                                                                                         |                                                       |
|---|---------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------|
| 1 | Performance & Speed | <i>AI-generated responses should be delivered quickly for a smooth user experience.</i> | <i>Response generated within <b>5–10 seconds</b>.</i> |
|---|---------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------|

| S.No | NFR Category | Requirement Description | Target Metric / Acceptance Criteria |
|------|--------------|-------------------------|-------------------------------------|
|------|--------------|-------------------------|-------------------------------------|

|          |                            |                                                                                                                                |                                                                                        |
|----------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>2</b> | Scalability                | <i>The application should support increasing numbers of users and AI requests without major performance degradation.</i>       | <i>Supports <b>multiple concurrent users</b> with stable response times.</i>           |
| <b>3</b> | Security & Data Privacy    | <i>Store API keys securely in .env files and never expose sensitive credentials to users or public repositories.</i>           | <i>API key never exposed; secure environment variable management.</i>                  |
| <b>4</b> | Reliability & Availability | <i>The application should remain stable during normal operation and gracefully handle invalid inputs or API failures.</i>      | <i><b>99% availability</b> during normal usage with meaningful error messages.</i>     |
| <b>5</b> | Usability & Accessibility  | <i>Provide a clean, responsive interface with clearly labeled fields and buttons for easy navigation.</i>                      | <i>Users can generate content in <b>three simple steps</b> with responsive design.</i> |
| <b>6</b> | Other                      | <i>The project should be modular, maintainable, and easy to extend with additional AI-powered content generation features.</i> | <i>Modular Flask codebase with reusable functions and clear project structure.</i>     |